

Session 8: FSDP and ZeRO

Trading Redundant Memory for Collective Communication

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Training Language Models: From First Principles to Efficient Serving

Lecture map

- 1 From replication to sharding
- 2 Execution and wrapping
- 3 Checkpointing and fair comparison

Guiding question

Which persistent states can be sharded, when must they be materialized, and what does that cost on the critical path?

What should be clear by the end?

- ① Account for DDP's replicated model-state memory.
- ② Explain ZeRO stages as progressive sharding.
- ③ Trace one FSDP unit through forward and backward.
- ④ Choose wrapping, precision, and checkpoint strategies from evidence.

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DDP replicates persistent state on every rank

State	bytes / param.	DDP placement
BF16 parameters	2	replicated
BF16 gradients	2	replicated
FP32 master copy	4	replicated
FP32 Adam moment m	4	replicated
FP32 Adam moment v	4	replicated
total model state	about 16	on every GPU

The scaling limit

More GPUs improve compute throughput but do not reduce DDP's replicated state.

ZeRO removes one class of redundancy at a time

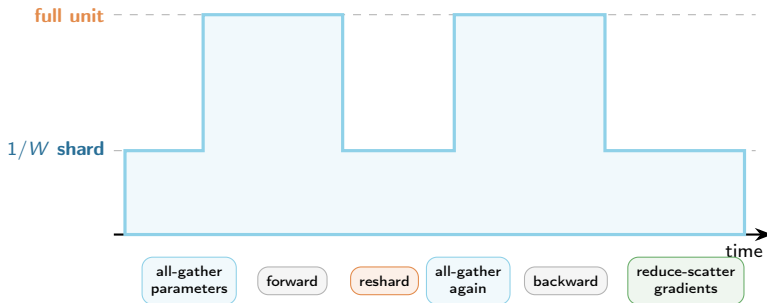
Method	optimizer state	gradients	parameters
DDP	replicated	replicated	replicated
ZeRO-1	sharded	replicated	replicated
ZeRO-2	sharded	sharded	replicated
ZeRO-3 / FSDP	sharded	sharded	sharded

Idealized per-rank model state

With full sharding and world size W , persistent model-state memory approaches $1/W$ of the unsharded total, before activations, buffers, and temporary materialization.

Source: Hugging Face Ultra-Scale Playbook and ZeRO

Sharding changes when full parameters exist



Full parameters are transient around compute; the local shard is persistent.

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One FSDP unit is a materialization boundary

Large unit

- fewer collectives
- larger peak all-gather
- less opportunity for overlap
- coarser memory release

Tiny units

- low peak materialization
- many latency-dominated collectives
- more metadata and scheduling overhead

Wrapping each Transformer block is a common starting point because it aligns materialization with a natural compute unit.

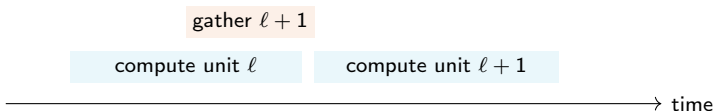
Communication volume and exposure are different

- **Volume:** total bytes transferred by all-gathers and reduce-scatters.
- **Exposure:** communication time that remains visible after overlap with compute.
- **Frequency:** number of collectives and their message sizes.

Why this matters

Two wrapping policies can move similar bytes but have very different step times because one creates many small latency-bound collectives or a longer exposed tail.

Prefetching tries to move the next unit early



Constraint

Prefetching improves overlap but raises transient memory because current and next full units may coexist.

FSDP does not shard activation memory automatically

Full sharding reduces persistent parameter, gradient, and optimizer-state redundancy.

Still scales with batch and sequence

saved activations
attention workspaces
temporary full parameters
communication buffers

Combine FSDP with activation checkpointing when activations, not model state, dominate the peak.

Mixed precision has three separate decisions

Quantity	Common choice	Main trade-off
parameter compute dtype	BF16	speed and temporary memory
gradient reduction dtype	BF16 or FP32	bandwidth versus numerical range
optimizer state dtype	FP32	stable moments at persistent cost

Experimental rule

When comparing DDP and FSDP, align precision policy. Otherwise, memory and numerical differences have multiple causes.

Checkpoint: wrapping policy

Think, then compare

A run fits in memory but launches hundreds of tiny all-gathers and scales poorly. What wrapping change would you test, and which memory and timing quantities could regress?

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A distributed checkpoint is a data layout

Full

- easy to consume
- expensive to gather
- rank 0 memory pressure

Sharded

- scalable writes
- world-size coupling
- unless resharded

Distributed format

- parallel I/O
- metadata and recovery
- complexity

Completion protocol

Publish a checkpoint only after every shard and its metadata are durable.
A directory that exists is not proof of a complete save.

A DDP–FSDP comparison needs a fixed contract

Match model, global token batch, optimizer, schedule, precision, data order, checkpoint cadence, and evaluation.

Correctness evidence

- first-update comparison
- loss trajectory
- resume equivalence

Systems evidence

- peak allocated memory
- tokens per second
- collective exposure
- checkpoint time and size

Choose FSDP when memory is the binding constraint

- Prefer DDP when the model fits comfortably and communication simplicity wins.
- Prefer FSDP when replicated state blocks model size, batch size, or context.
- Tune wrapping and prefetch from traces, not from a universal recipe.
- Treat checkpoint load and resharding as part of the production design.

Checkpoint: memory prediction

Think, then compare

Estimate idealized model-state memory per rank for P parameters, 16 bytes per parameter, and world size W . Then name three reasons the measured peak will be higher.

The durable ideas

- 1 ZeRO stages progressively shard optimizer state, gradients, and parameters.
- 2 FSDP materializes one wrapped unit through all-gather and returns sharded gradients.
- 3 Wrapping controls peak memory, collective size, frequency, and overlap.
- 4 Checkpoint format and a matched DDP baseline are part of the system claim.